

Requirements-based testing of deep neural networks.

RBT4DNN · A critical review of its empirical evidence and two focused additional analyses.

SEMINAR

AI Testing Landscape: Ensuring Robustness through Advanced Testing Methods

COURSE LEADERSHIP

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PRESENTATION

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Three questions guide the argument.

01 • NEED

Why do neural-
network tests need
requirements?

PROBLEM + BACKGROUND

02 • METHOD

How does RBT4DNN
generate and evaluate
targeted tests?

APPROACH + METHODOLOGY + RESULTS

03 • EVIDENCE

How convincing—and
scalable—is the
resulting evidence?

CRITIQUE + ADDITIONAL ANALYSES

High average accuracy does not answer a specific requirement.

CONVENTIONAL TESTING Known behavior A specification maps an input condition to an expected output. Test oracles can check the result directly.	DNN TESTING Learned behavior The model is inferred from examples. Global accuracy summarizes a distribution, not every meaningful scenario.	THE MISSING LINK Semantic scenarios Developers need evidence for statements such as “a very thick 7 must still be classified as 7.”
GLOBAL METRIC How often is the complete test set correct?	→ SEMANTIC SLICE Which rare inputs does the score contain?	→ REQUIREMENT VERDICT Does the model satisfy this specific contract?

[1] §1-§2.5 · “model under test” means the evaluated neural network

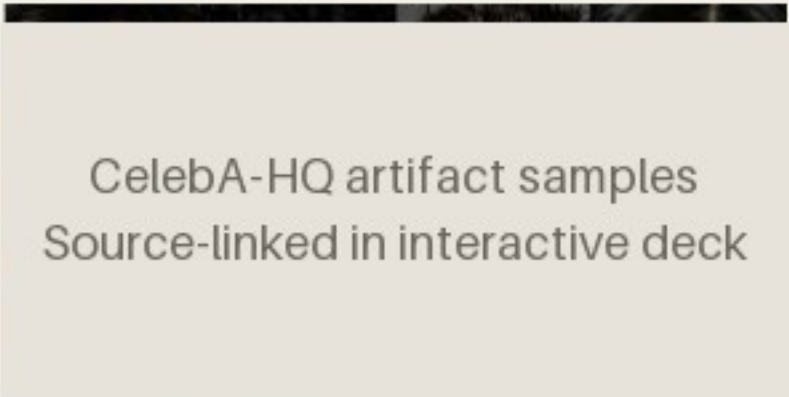
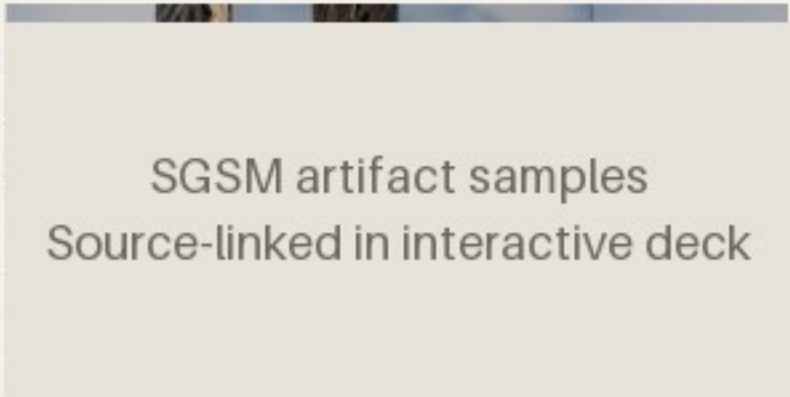
Existing generators rarely start from the requirement itself.

Neuron coverage, image transformations, and search-based generators can produce many tests—without proving that those tests represent the scenario a stakeholder cares about.

INPUT	Requirement	Structured natural language defines a semantic precondition and an expected outcome.
METHOD	RBT4DNN	Fine-tune a generator on images that satisfy that precondition.
OUTPUT	Targeted tests	Run the learned component and evaluate the requirement-specific oracle.

[1] §1, §2.5 and §3 - authors claim the first requirement-driven DNN test-input generator

Four domains test whether one workflow survives increasingly ambiguous semantics.

MNIST · 7 REQUIREMENTS  MNIST artifact samples Source-linked in interactive deck MNIST Measured height, thickness, and lean. A controlled starting point. Output: digit class	CELEBA-HQ · 7 REQUIREMENTS  CelebA-HQ artifact samples Source-linked in interactive deck CelebA-HQ Ambiguous attributes labeled by visual question answering or humans. Output: eyeglasses decision	SGSM · 7 REQUIREMENTS  SGSM artifact samples Source-linked in interactive deck SGSM Synthetic driving scenes with spatial relations linked to steering and acceleration. Output: numeric control	IMAGENET · 4 REQUIREMENTS  ImageNet artifact samples Source-linked in interactive deck ImageNet Animal morphology linked to allowed classes in the WordNet taxonomy. Output: class set
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[1] Table 2 and §4.1.1-§4.1.3 · samples linked from the pinned public artifact

MNIST M3 artifact samples
Source-linked in interactive deck

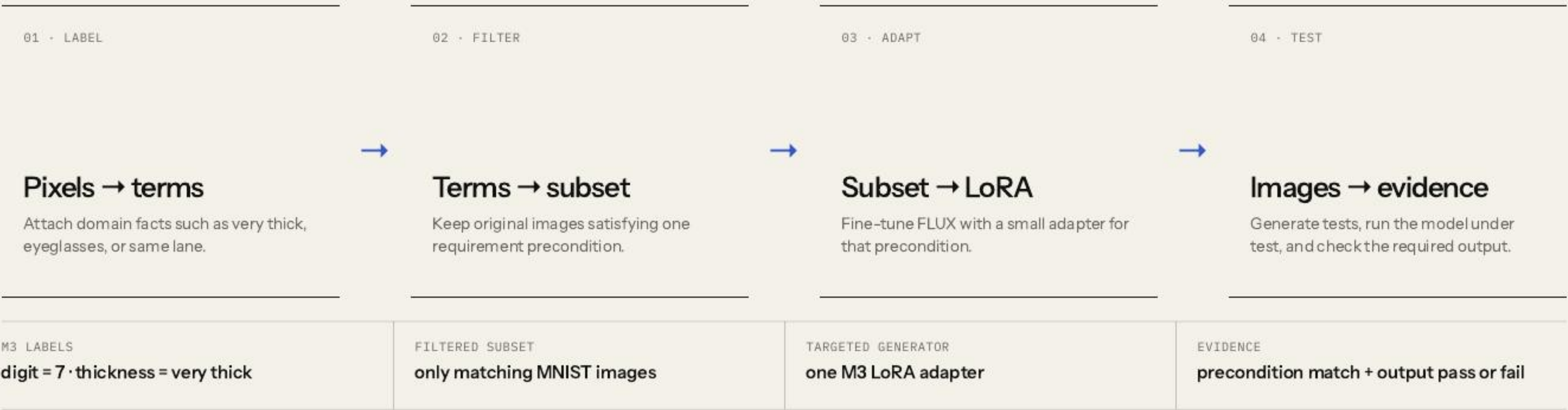
A requirement defines both relevance and correctness.

If the digit is a 7 and is very thick, then the model shall label it as 7.

PRECONDITION	POSTCONDITION
Which images count as relevant tests?	Which model outputs count as acceptable?

[1] Table 2, M3 · a finite passing suite cannot prove the universal requirement

RBT4DNN turns one requirement into a targeted test campaign.



[1] Fig. 2 and §3 · symbols are translated into an explicit input/output contract

Glossary terms make visual requirements operational.

GLOSSARY TERM

A named visual property that can be assigned to an image: “very thick,” “same lane,” or “eyeglasses.”

01 · DEFINE

Authors specify the glossary

Requirement concepts are manually turned into a finite vocabulary of visual properties.

TERM CREATION IS NOT AUTOMATIC

02 · LABEL ORIGINAL DATA

Labels come from domain-specific signals

MNIST measurements · SGSM scene graphs · MiniCPM visual Q&A for CelebA-HQ and ImageNet.

CELEBA ALSO HAS HUMAN LABELS

03 · USE THE LABELS

Filter first; check later

Labels select the LoRA training subset. Separate binary classifiers then check glossary terms on generated images.

PRECONDITION MATCH · RQ1 / RQ3

Boundary: separate training labels and evaluation classifiers reduce direct reuse, but neither constitutes independent semantic ground truth.

[1] §3.2 and §4.1.6 · mean glossary-classifier accuracy 94.5%; C4 is 70.8%

One filtered precondition subset trains one targeted LoRA.



BENEFIT	Generate many new candidates inside a rare requirement region
EVIDENCE SOUGHT	Relevance, distributional plausibility, and semantic variation
NOT ESTABLISHED	Non-memorization or systematic coverage within that region

[1] §3.3 and §4.1.4 - LoRA equation simplified; latent-space coverage remains future work

The six research questions validate inputs before interpreting model outcomes.

A **credible failure is a joint event**: the generated input satisfies the precondition, but the model output violates the postcondition. RQ = Research Question.

Validate the generated inputs

RQ1-RQ3

- RQ1

Consistency

Does the image satisfy the precondition?
- RQ2

Plausibility

Is its distribution close to matching training images?
- RQ3

Variation

Are other glossary features distributed similarly?

Evaluate the resulting tests

RQ4-RQ6

- RQ4

Comparison

Are targeted outcomes more informative than baselines?
- RQ5

Violations

Do tests reveal credible postcondition failures?
- RQ6

Diagnosis

Do failures expose repeatable decision patterns?

[1] §4.1 · input validity is established before model outcomes are interpreted

Targeted LoRAs satisfy the requested precondition 92.1% of the time on average.

92.1%

mean across 21 requirements in MNIST, CelebA-HQ, and SGSM.

M3 · targeted LoRA

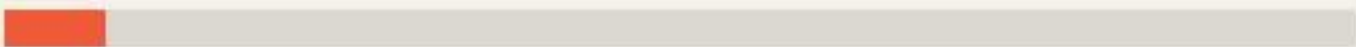
filtered images: very thick 7



95.8%

M3 · one all-data LoRA

all annotated MNIST training images; not plain FLUX



7.4%

COMPARISON SHOWN

Precondition-specific LoRA versus one LoRA trained on all annotated MNIST data · 100 images × 10 repetitions.

BOUNDARY

Glossary classifiers average 94.5% accuracy; they are measurement proxies, not semantic ground truth.

KID compares each generator with real images satisfying the same precondition.

MNIST · M3 · KID ↓ .308 → .039 PROMPT-ONLY FLUX TARGETED LORA .308 .039	SGSM · S2 · KID ↓ .242 → .038 PROMPT-ONLY FLUX TARGETED LORA .242 .038	CELEBA-HQ · C1 · KID ↓ .123 → .097 PROMPT-ONLY FLUX TARGETED LORA .123 .097
KID · KERNEL INCEPTION DISTANCE A distance between two image sets in a learned feature space · lower means the distributions are closer.	BOUNDARY KID is not an image-level realism probability; several reference subsets contain fewer than 300 images.	

[1] §4.2.2 and Table 3 · examples show KID(reference, prompt-only FLUX) → KID(reference, targeted LoRA)

Against the same filtered training subset, targeted LoRA preserves the feature mix better than prompt-only FLUX.

MNIST · MEAN JS 7.2× closer PROMPT-ONLY FLUX TARGETED LORA .1765 .0244	CELEBA-HQ · MEAN JS 1.4× closer PROMPT-ONLY FLUX TARGETED LORA .1150 .0831	SGSM · MEAN JS 5.5× closer PROMPT-ONLY FLUX TARGETED LORA .1960 .0355
WHAT “FEATURE FREQUENCY” MEANS Example: among thick 7s, how often are images also left-leaning, right-leaning, tall, or short? Compare those label proportions with the filtered training subset.	JS · JENSEN-SHANNON DIVERGENCE One distance between those two label distributions · lower means closer. It does not measure pixel diversity, memorization, or latent-space coverage.	

[1] §4.2.3 and Table 4 · displayed domain means are calculated from the seven reported rows

M2 requirement: if the image is a **very thick 3**, the classifier must predict 3.

Technique	Passed Tests	Failed Tests
Deephyperion		
Image Transform		
RBT4DNN		

Original paper evidence · Table 5 · cropped only for legibility · CC BY 4.0.

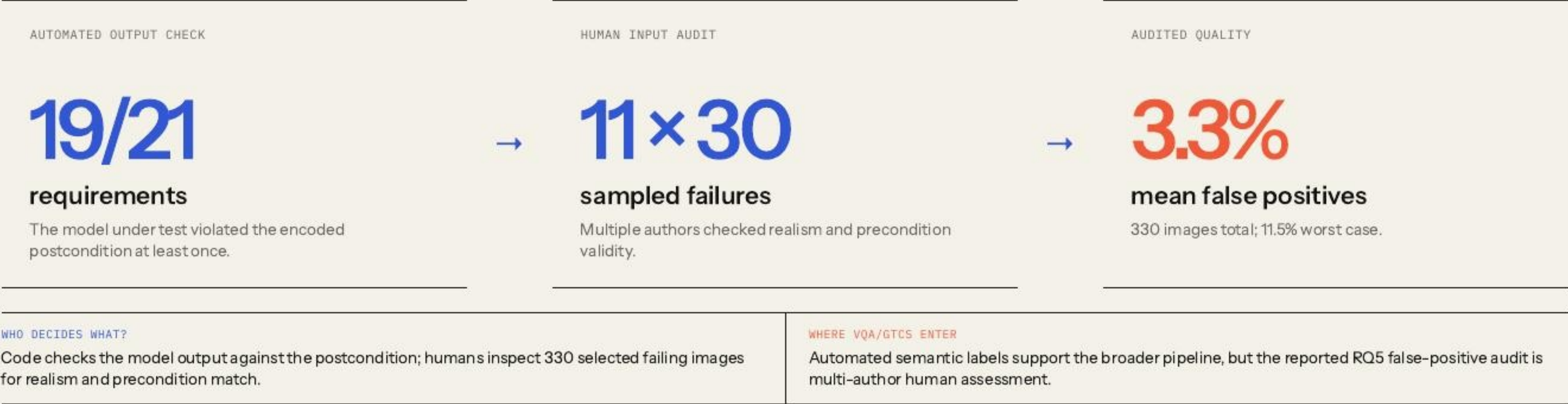
WHAT PASS / FAIL MEANS HERE

For M2, pass = the MNIST classifier predicts “3”; fail = it predicts another digit. The rows separate those outputs for each generator.

INTERPRETATION RULE

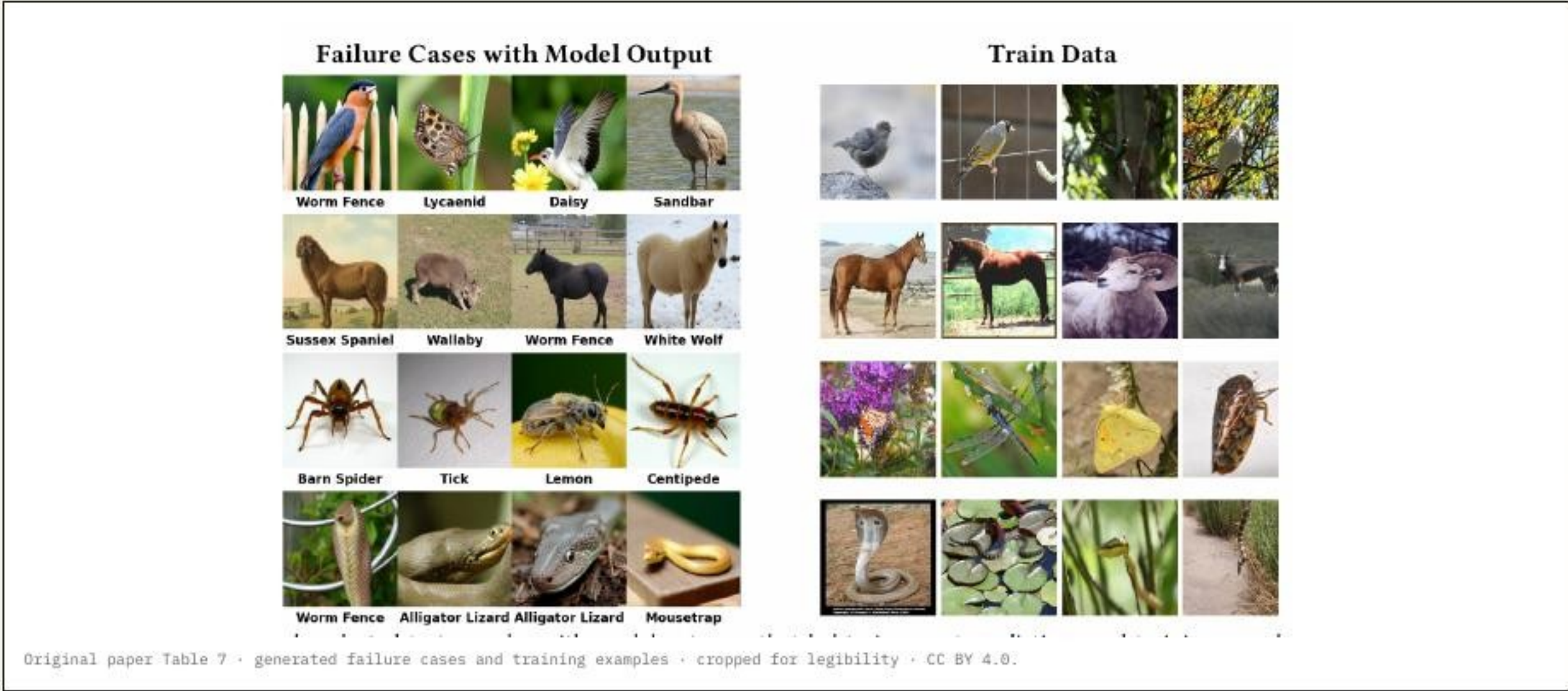
DeepHyperion and transformations mostly miss “very thick”; therefore neither their passes nor failures are evidence about M2.

RBT4DNN exposed failures broadly, and the audited subset was mostly valid.



[1] §4.1.6 and §4.3.2 · assessor disagreement on precondition consistency was 1.7%

Across four animal requirements and three classifiers, failures reveal recurring model-specific patterns.



METHOD

4 requirements × 3 classifiers × repeated generation

Collect 1,486 postcondition failures, rank frequent wrong labels, then manually inspect associated images.

EXAMPLE

A bird near a fence → “worm fence”

Other cases align with insects, horses, or snakes that resemble frequent training classes or backgrounds.

BOUNDARY

Association, not causal diagnosis

The study does not change only the background while holding the animal fixed.

The paper demonstrates feasibility—not a complete assurance process.

GENERATIVE TRAINING Better tests Improve generator training and adaptation to reveal more observed failures and reduce false positives.	COVERAGE EVIDENCE Latent-space coverage Map the model's internal feature space and deliberately search regions that generated tests have not exercised.	GENERALIZATION More domains Expand driving-like safety datasets and explore broader robustness and functional requirements.
ASSURANCE BOUNDARY The comparator changes with the question; automated labels, narrow baselines, and small KID subsets limit how far the evidence generalizes.		

[1] §4.4 and §5 · author-stated scope kept separate from this seminar's critique

A model error counts as requirement evidence only if the generated image really satisfies the requirement.

PRECONDITION INVALID · MODEL PASSES Irrelevant pass The output says nothing about the stated requirement.	PRECONDITION INVALID · MODEL FAILS Off-target false positive A model error is not yet a requirement counterexample.
PRECONDITION VALID · MODEL PASSES Valid requirement pass Relevant evidence, but never proof of the universal property.	PRECONDITION VALID · MODEL FAILS Credible counterexample This joint event is the paper’s strongest testing evidence.

SAMPLED JOINT AUDIT

30 failures × 11 low-pass requirements = 330 independently assessed samples.

Critique in one sentence: the authors manually verify 330 selected failures, but most reported results still depend on imperfect automated labels—so confidence is strongest for the audited sample, not the full failure population.

Sharing one MNIST LoRA saves training—but sharply weakens targeting on the hardest requirements.

METHOD	EASY TO SHARE · M4	HARD TO SHARE · M3
Reuse the authors' MNIST ablations Compare precondition-match rates for a targeted LoRA, one LoRA over six requirement subsets, and one LoRA over all MNIST data. 6 requirements	Targeted 92.5% · shared 56.4% The shared requirement LoRA remains usable, though substantially less precise. The all-data LoRA reaches only 12.7%. training ↘ · targeting ↘	Targeted 95.8% · shared 27.5% The all-data LoRA reaches 7.4%. A single generator does not preserve difficult semantic regions automatically. targeting matters

EXTENSION MESSAGE

Per-requirement LoRAs are not merely an implementation detail: they buy semantic precision. The open design question is whether conditioning can recover that precision in one shared model.

Sharing can cut assumed training compute by 75%—but loses the targeting that makes tests relevant.

6 TARGETED LORAS One adapter per requirement Six assumed 1-hour training runs. Each adapter sees only images matching its own precondition. \$18 · M3 95.8%	1 SHARED-REQUIREMENTS LORA Union of six filtered subsets One assumed 1.5-hour run. Cheaper, but the requirement signal competes inside one adapter. \$4.50 · M3 27.5%	1 ALL-DATA LORA All annotated MNIST data One assumed 1.5-hour run. Cheapest model count, but no requirement-specific filtering. \$4.50 · M3 7.4%
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WHAT IS OURS—AND WHAT IS ASSUMED

Targeting rates come from the authors' saved ablations. Costs are a seminar scenario at \$3/GPU-hour—not invoices or newly trained LoRAs. The extension makes the scalability–precision trade-off explicit.

Requirements make tests **relevant**. Validity makes failures **credible**.

WHAT THE PAPER ESTABLISHES

Structured requirements can guide distributionally plausible, glossary-diverse, scenario-specific test generation across four domains.

WHAT LIMITS THE CLAIM

Semantic labels and glossary classifiers are useful proxies; credible failures still depend on joint input-and-output validation.

References.

[1]

N. J. Mozumder, F. Toledo, S. Dola, and M. B. Dwyer. “RBT4DNN: Requirements-based Testing of Neural Networks.” arXiv:2504.02737, v3, 2025. CC BY 4.0.

[2]

E. J. Hu et al. “LoRA: Low-Rank Adaptation of Large Language Models.” ICLR, 2022.

[3]

RBT4DNN artifact. github.com/less-lab-uva/RBT4DNN · seminar provenance pinned to commit c3deff871ed41043d83a05289faab84c41706586.

[4]

Seminar experiments. Reproducible notebooks, results, summaries, and provenance in this repository.

Original paper ↗	Original artifact ↗
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